

Objectives

- The objective of this lab is to apply the concepts learned in class about diodes and learn about other useful applications of diodes.

Equipment

- 1N4148
- 1N5817
- BZX55B3V9-TR
- Breadboard and various wires
- Various resistors
- Digilent Analog Discovery (DAD) Board
- LTspice and Waveforms

Introduction

Diodes are a semiconductor junction device that conducts current in one direction. They are found in everything from power supplies and rectifiers to signal modulation and light-emitting displays. Their unidirectional current flow, combined with a well-defined forward voltage threshold and the ability to withstand reverse breakdown voltages, allows them to function as rectifying elements in AC-to-DC converters, switching gates in digital logic, and voltage clamps that protect circuits from transients. More specialized variants, such as Zener diodes, light-emitting diodes (LEDs), and Schottky diodes, extend this versatility further. Zener diodes stabilize supply rails through controlled reverse breakdown, LEDs provide low-power illumination and communication, and Schottky diodes offer ultra-fast, low-forward-voltage switching essential for high-speed signal paths.

Often, we want to make a voltage that will remain constant even if the circuit conditions change.

For the example below: Examine the circuit below intended to provide 25 mA to charge a battery.

Assume $R_1 = 79 \text{ k}\Omega$; $R_2 = 1 \text{ k}\Omega$; $R_{\text{sense}} = 10 \text{ }\Omega$; $V_{\text{battery}} = 3.7 \text{ V}$. If V_{DD} changes from 20V to 30V, the current becomes 37.5 mA. This can be problematic if we need the current to remain at a certain level.

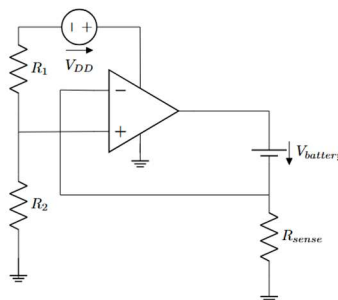


Figure 1- Diode Example I

Lab 3: Diode Applications – Pre-Lab

This supply dependence can be quite problematic, especially if the current needs to be precisely controlled. In class we learned about diodes and how they have the ability to keep a roughly constant voltage across their terminals even when the current through them changes drastically. We will use this property to build a precision voltage reference.

Pre-Lab

Part 1: IV Curve & Operating Point (20 Points)

- I. Download the datasheet for the 1N4148 switching diode. Write down the maximum continuous reverse voltage, continuous forward current, and power dissipation ratings for your own reference. You should be aware of these limits when doing the experiments. **(2 Points)**
 - a. **Maximum Continuous Reverse Voltage:** 75V
 - b. **Continuous Forward Current:** 300mA
 - c. **Power Dissipation:** 500mW for $T \leq 25^\circ\text{C}$ or 440mW for $T = 45^\circ\text{C}$
- II. Build the circuit below in LTSpice to characterize the IV characteristics of the 1N4148 diode. Right click on the diode and press the “pick diode” button on the pop-up to choose the 1N4148. **(3 Points)**

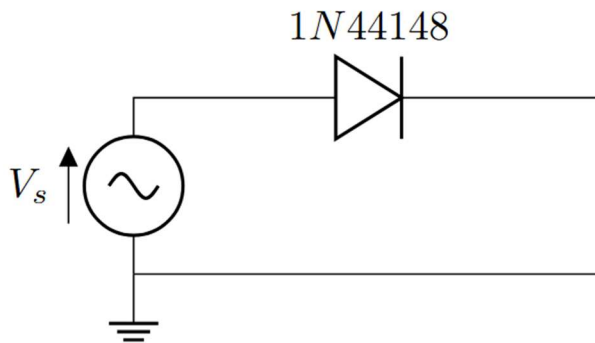
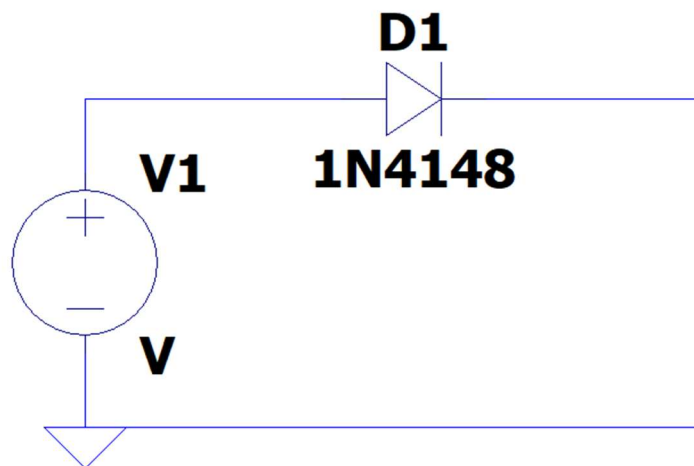
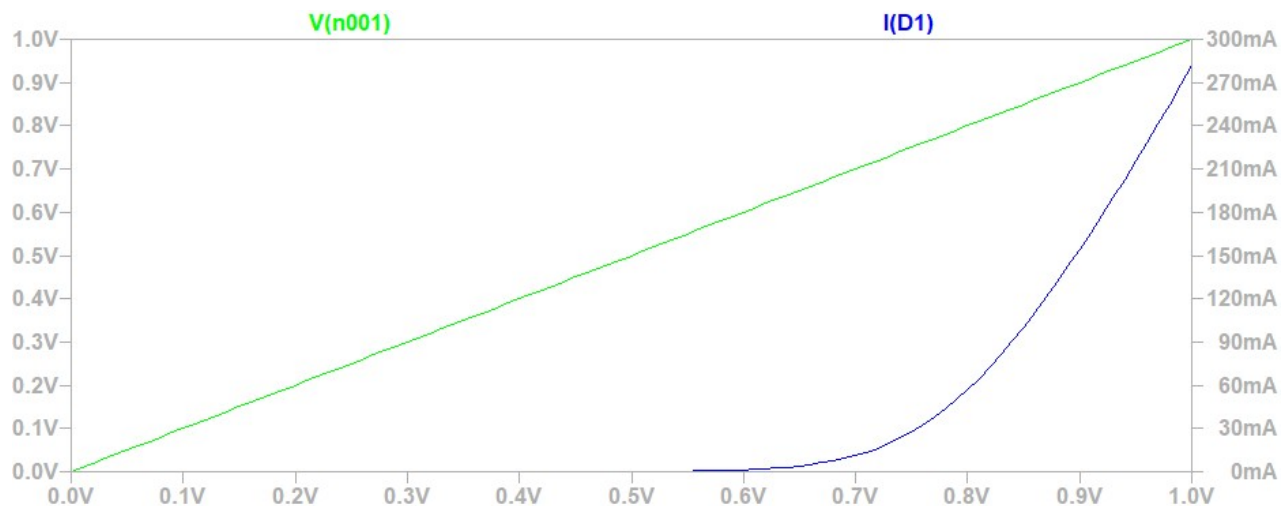


Figure 2 – Diode Example II



- III. Plot the IV curve for the diode and include it below. *Do this by running a DC sweep from 0 to 1V. **(3 Points)**



- IV. Using the IV curve, find the slope when $V_{\text{diode}} = 0.7 \text{ V}$ & $V_{\text{diode}} = 1 \text{ V}$. Report the slope below along with the current for both voltages. *For the slope, use 2 close points + Rise/Run. What are the units of the slope? **(2 Points)**

Point 1 (0.7V): 700.02448mV and 11.203113mA

Point 2 (0.7V): 700.56304 mV and 11.330113mA

Slope: 0.23581 mA/V

Point 1 (1V): 999.46144mV and 281.2464mA

Point 2 (1V): 1V and 281.98141mA

Slope: 1.3647mA/V

- V. Suppose we apply the following signal that varies between 0.71 V and 0.69 V across the diode. Approximate the current through the diode? **(5 Points)**

$$\text{Signal: } v(t) = 0.7 + 0.01\sin(\omega t)$$

*Assume that the slope of the IV cure doesn't change much within $\pm 10\text{mV}$ of 0.7V

$$11.203113 \pm 0.23581 \cdot 0.01 = 11.2007549, 11.2054711 \text{ mA}$$

Note: If we assume signals are small enough then we can approximate the IV characteristics of the diode with a resistor. We can find that resistance using an IV curve. This is done by dividing the voltage by the current at the point where exponential growth begins.

VI. Which of the two operating point gives a smaller “small signal” diode resistance: 0.7V or 1V? **(3 Points)**

$R = 1/(mA/V)$:

- for 0.7V $\rightarrow 1/(0.23581 * 10^3) = 4240 \text{ Ohm}$
- for 1V $\rightarrow 1/(1.3648 * 10^3) = 732.7 \text{ Ohm}$

VII. What is the power dissipated by the diode for the two points? **(2 Points)**

$P = VI$:

- For 0.7V $\rightarrow .70002448 * 11.284572 * 10^{-3} = 7.90 \text{ mW}$
- For 1V $\rightarrow 1 * .28198141 = 282 \text{ mW}$

Part 2: Biasing to a Specific Operating Point (10 Points)

The circuit in Figure 3 takes advantage of the fact that the diode voltage is roughly constant to apply a specific current through it. We can set an operating point by inspecting the IV curve for the diode and assuming the voltage and current across the diode.

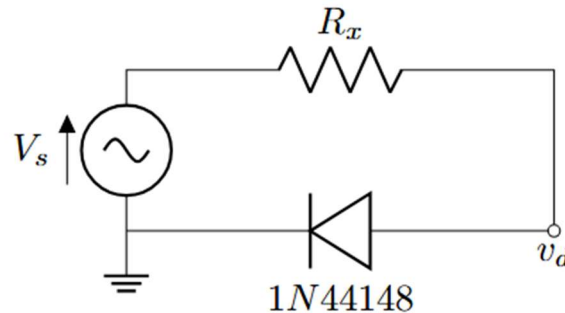


Figure 3 – Diode Example III

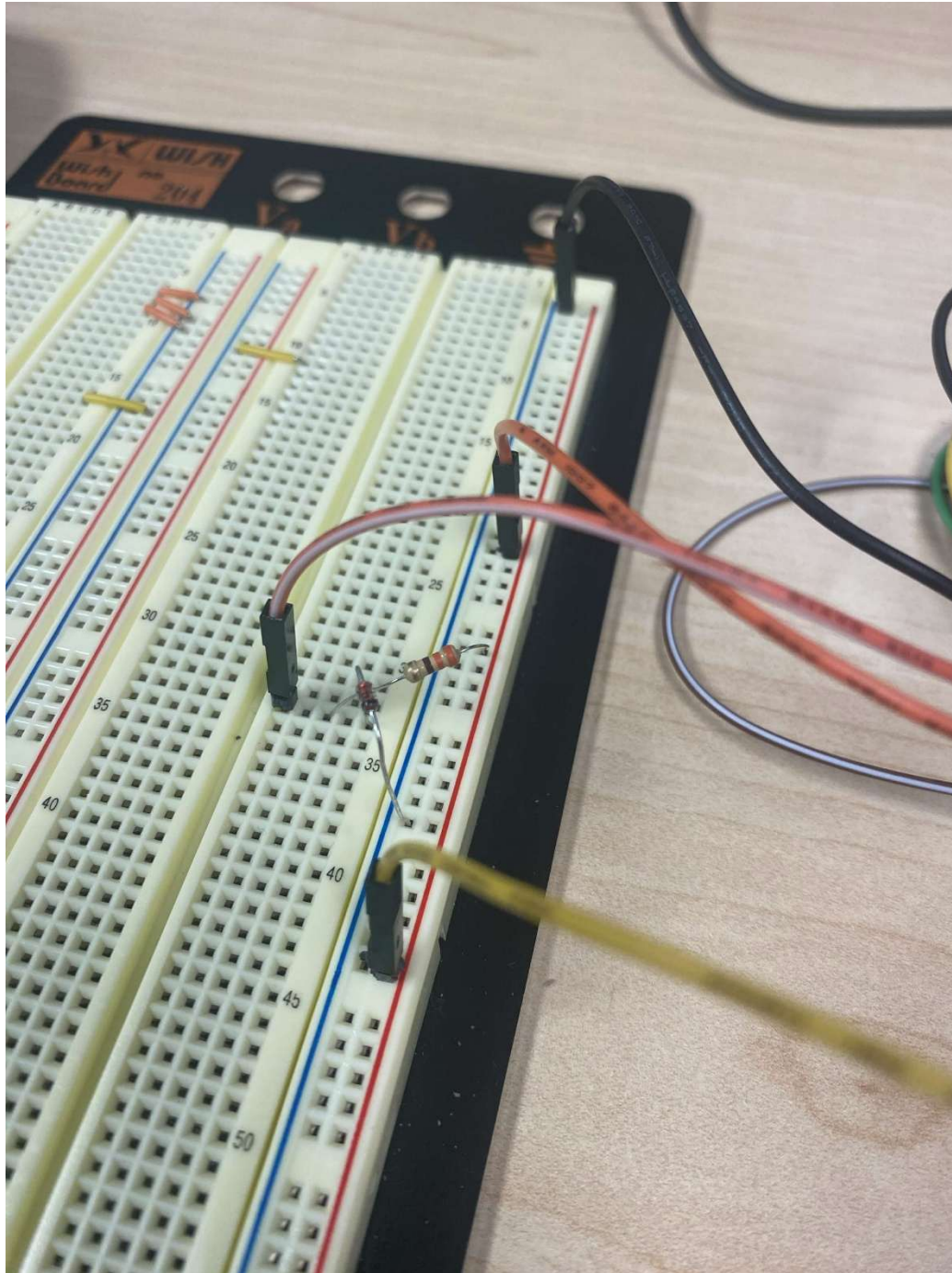
- I. Assume that the diode voltage and current are 0.7 V and 11 mA respectively. If our supply voltage was 4 V DC, find the R_x to ensure that V_d and I_d are 0.7V and 11mA. **(5 Points)**

$$4 - 0.7 = 3.3\text{V}$$

$$3.3\text{V}/11\text{mA} = 300\Omega$$

*Chances are you don't have a resistor of that exact value but if you are within 10% of the calculated resistance value then V_{diode} will be within 5 mV of 0.7 V.

- II. Complete the following items:
- (1) Build the circuit on your bread board.
 - (2) Provide a picture of your circuit.



- (a)
- (3) Find V_{diode} using waveforms or a multimeter. State the (was blank, V?).
(a) I got .731V across the diode, a bit high, unsure why.
 - (4) Repeat for I_d .
(a) $4 - .731 = 3.269\text{V} \rightarrow 3.269/330 = 9.9\text{mA}$
 - (5) Note the resistor value you used for R_x .
(a) 330 Ohm Resistor

(5 Points)

Part 3: Half-wave Rectifier (10 Points)

Half-wave rectifiers are the simplest form of DC power conversion, taking an AC voltage and “cutting away” one half of each cycle so that only the pulses of one polarity reach the load. During the positive half-cycle of the AC input, the diode forward-biases, allowing current to flow through the load. The instantaneous output voltage equals the input minus the diode’s forward drop. In the negative half-cycle the diode blocks current, so the load sees zero voltage. The result is a series of positive spikes whose amplitude is limited by the input peak and the diode drop.

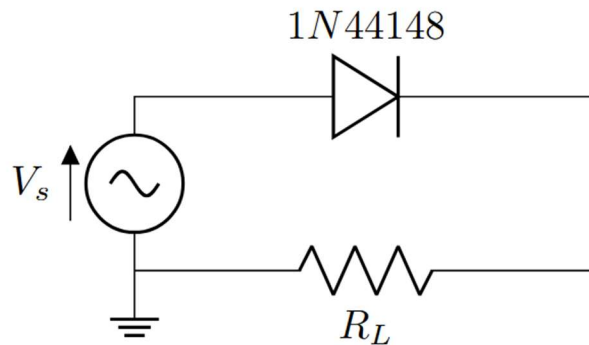
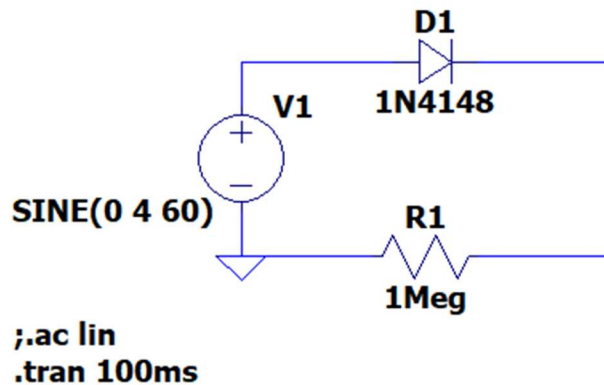
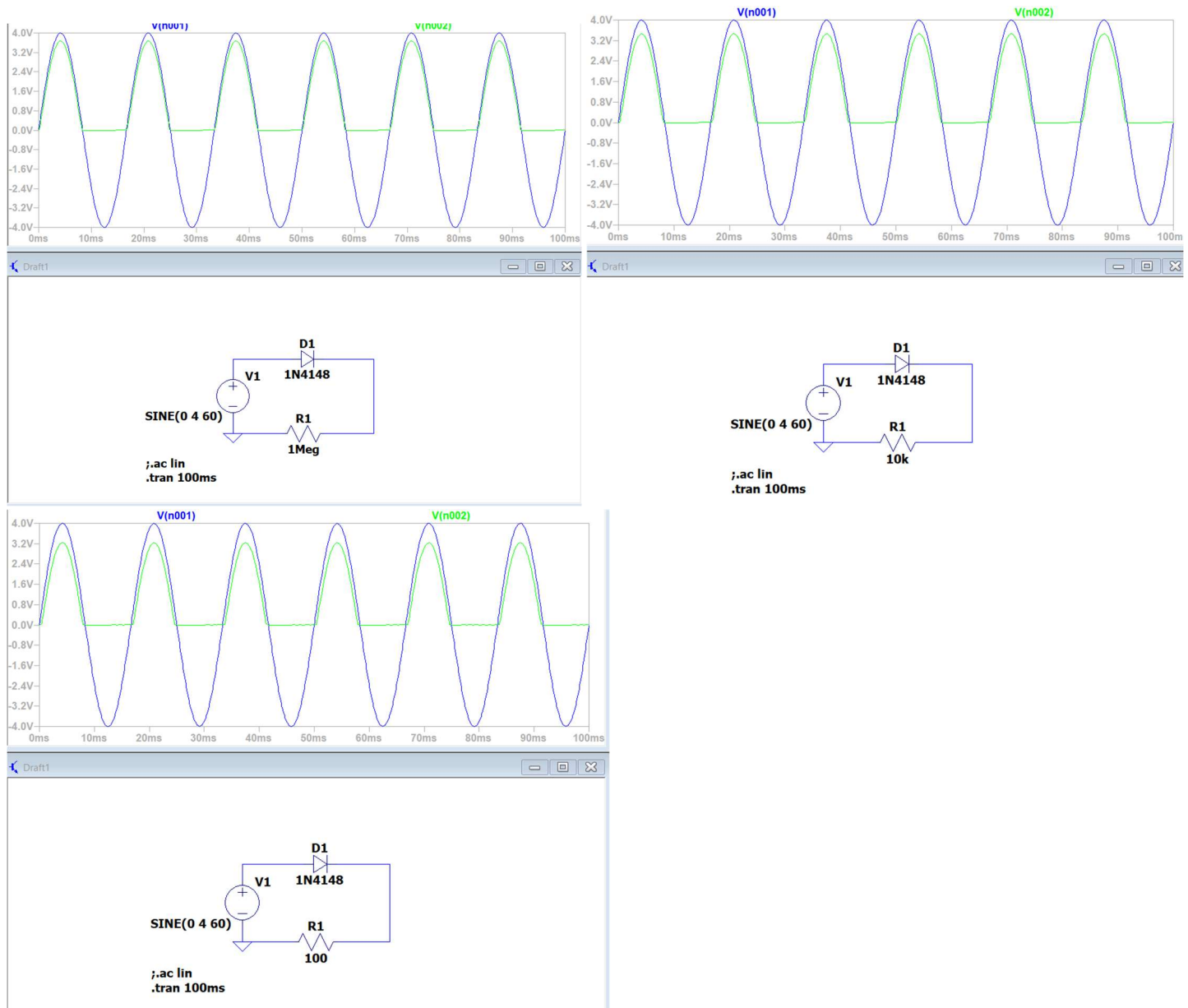


Figure 4 - Half-wave Rectifier Circuit

- I. Construct the following circuit in LTspice. (4 Points)



- II. Experiment with various values of R_L to see how it affects the outputs behavior across the load. Explain your results. Include at least three figures to explain your findings. **(4 Points)**



As R_L increases in size, the voltage drop for the positive half of our input decreases across the diode.

- III. Think of one application in which this device may be useful. **(2 Points)**
 Any application where you want it to have a pulse of energy, such as a light on the top of a skyscraper to direct planes at night.

Part 4: Clippers (20 Points)

As the name implies, a diode “clipper” circuit will be one intended to remove, or “clip off,” part of the output waveform. These may be used singly (Figure 5) to prevent reaching a certain value in the positive or negative portion of a waveform, or they can be used in pairs (Figure 6) to perform double-ended limiting at two independent levels. *R = 10 kΩ

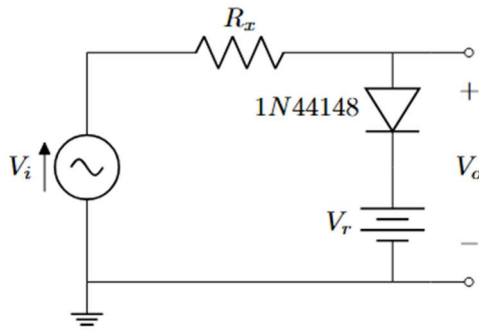


Figure 5 - Single Diode Clipper Circuit

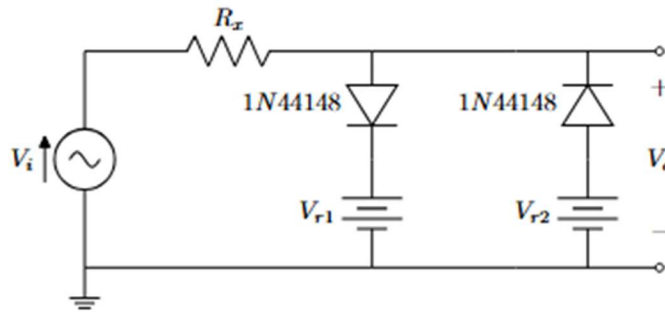
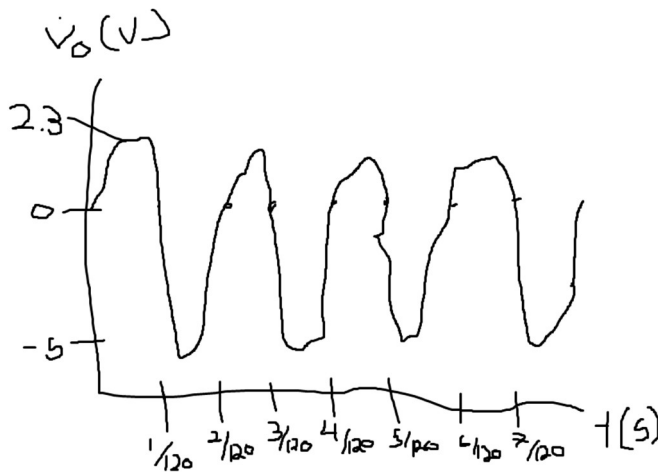
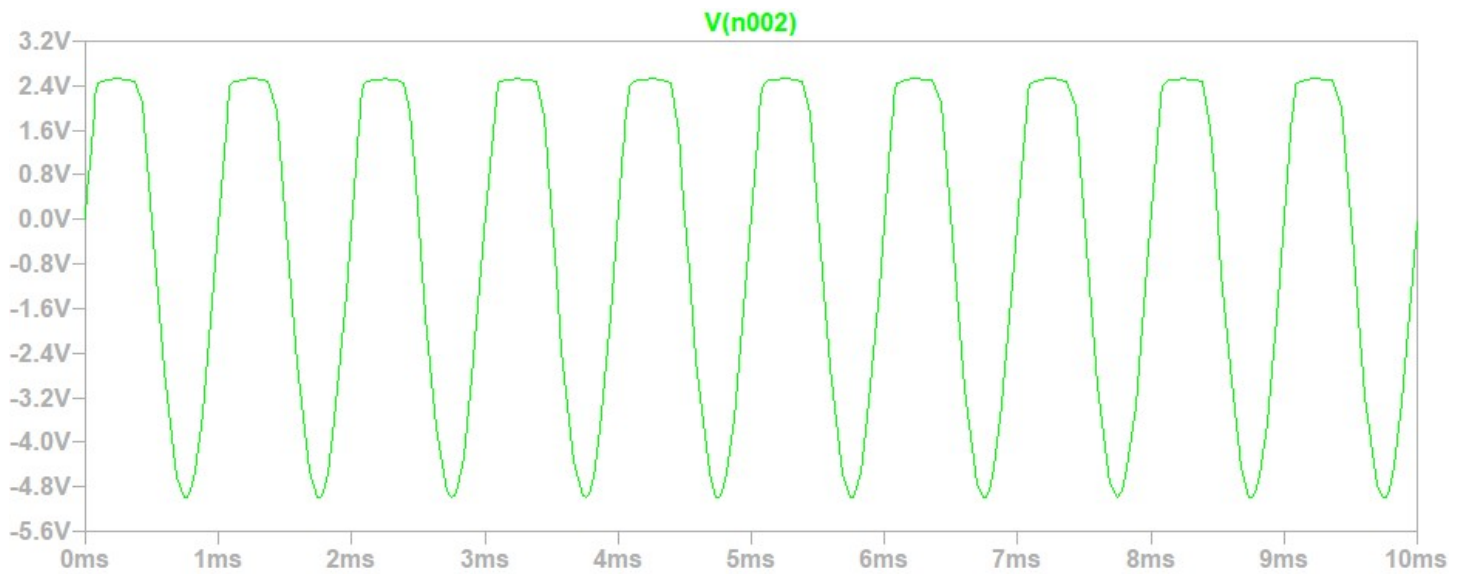


Figure 6 - Double Diode Clipper Circuit

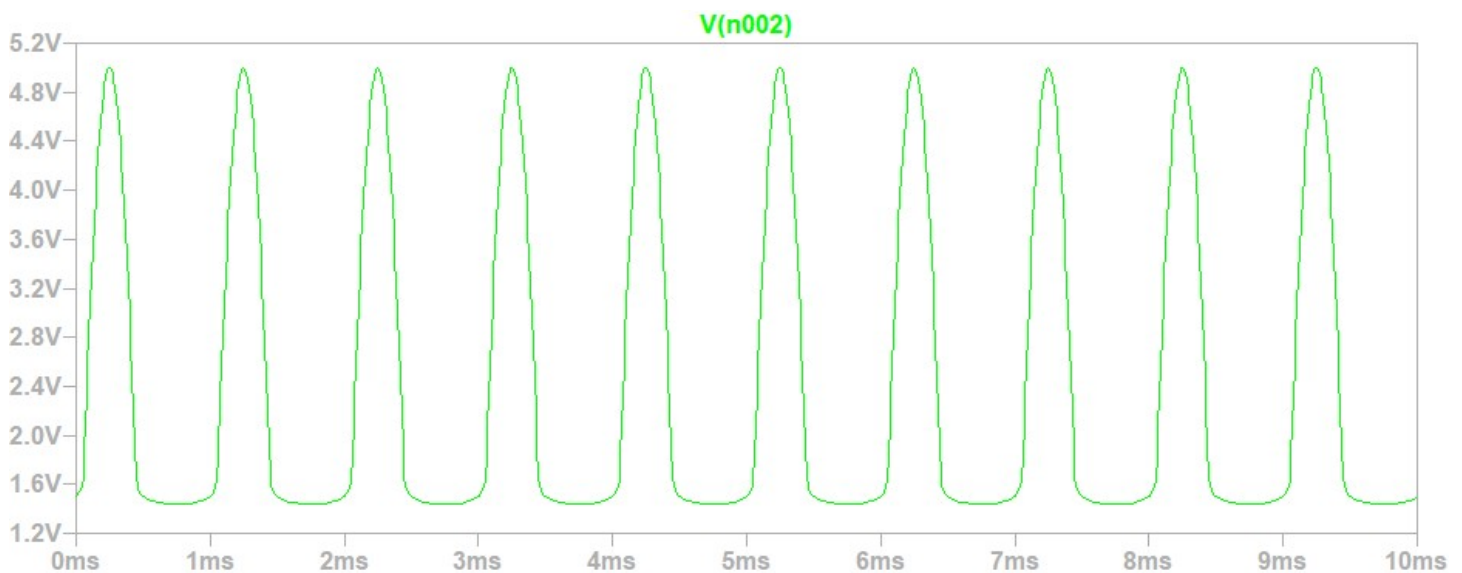
- I. Consider the circuit shown in Figure 5. Let $V_i(t) = 5.0 \cdot \sin(377 \cdot t)$ and $V_r = 2.0\text{V}$. Sketch what you expect the output waveform $V_o(t)$ to be and show the maximum and minimum values of the output voltage. (1 Points)



- II. Simulate the clipping circuit shown in Figure 5. Let $V_r = 2\text{V}$ and the input voltage V_i be a 5 V-peak, 1 kHz sinusoidal voltage. Connect an oscilloscope (in Multisim) to V_o and adjust the timescales and volts/division as necessary. Take a screenshot of this waveform and compare it to your sketch. (4 Points)

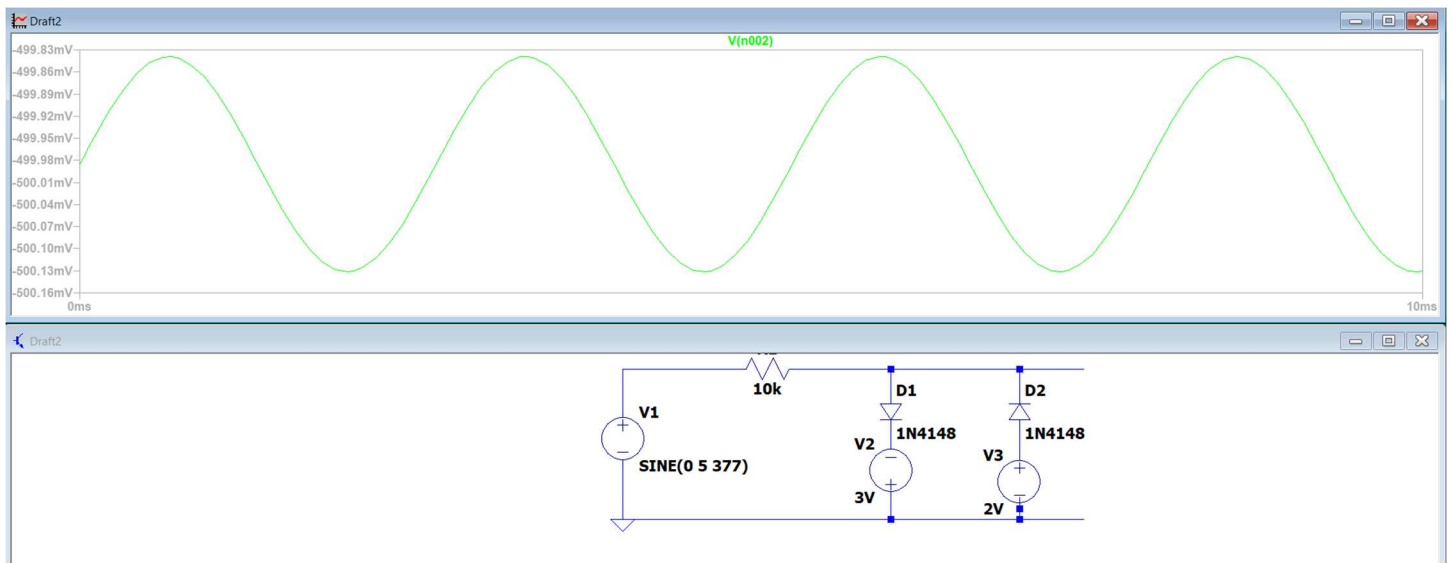


- III. Reverse the direction of the diode in your simulation. Observe the output voltage v_o waveform on your oscilloscope. How does this voltage compare to that in the previous step? **(2 Points)**



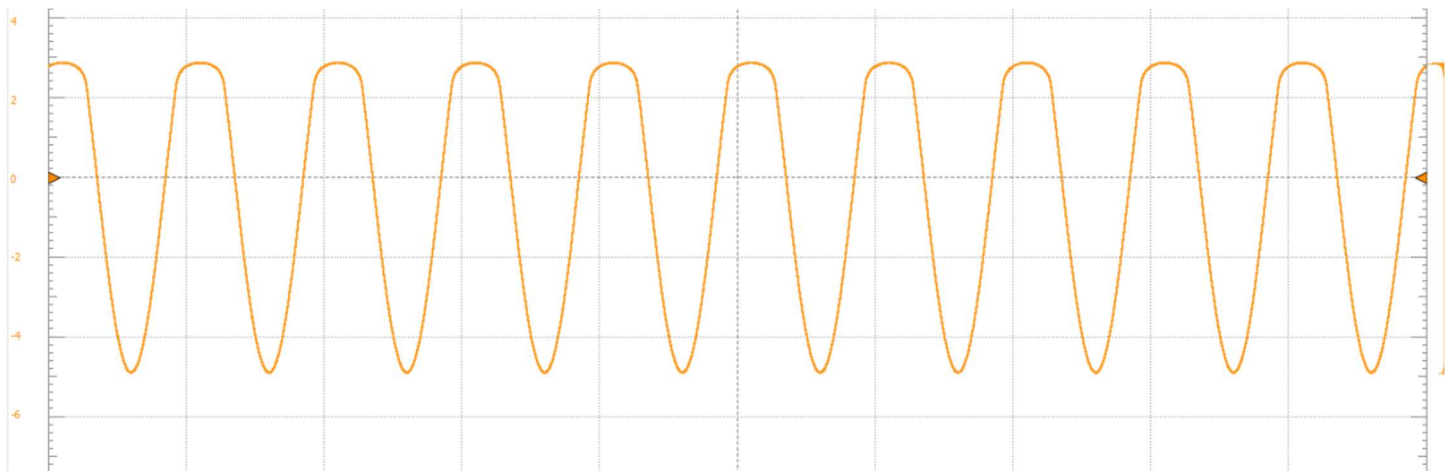
Rectifies opposite edge

- IV. Consider the circuit shown in Figure 6. Let $V_i(t) = 5.0 \sin(377t)$, $V_{r1} = 3.0$ V, and $V_{r2} = 2.0$ V. What is generally different about this waveform compared to that of the previous parts? **(4 Points)**

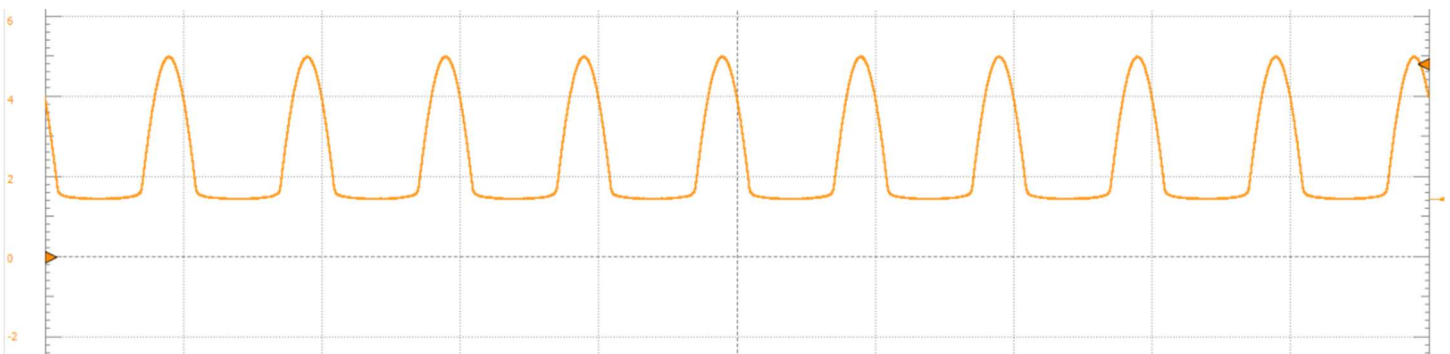


Smooth curve from equal diode effect on both halves

- V. Construct the clipping circuit shown in Figure 5. Let $V_R = 2\text{ V}$ and the input voltage v_i be a 5-peak, 1 kHz sinusoidal voltage. Observe the output voltage V_o . This circuit transmits that part of the V_i waveform that is more negative than $V_r + V_T$. **(3 Points)**

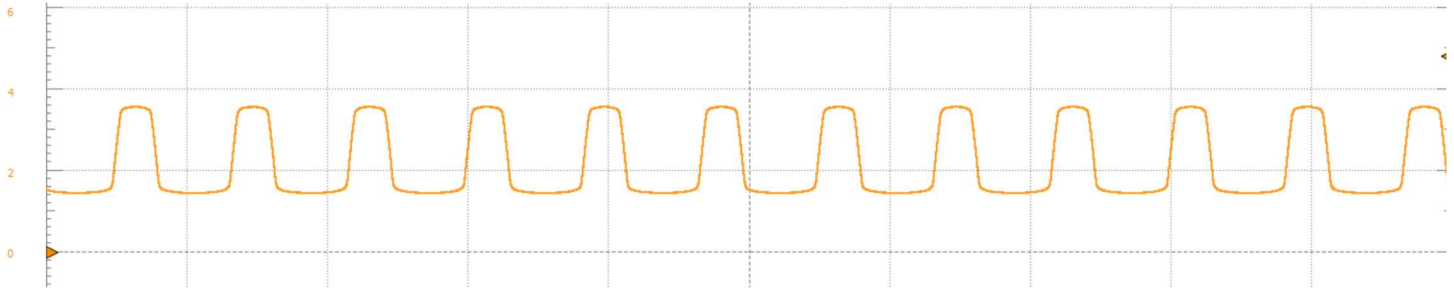


- VI. Reverse the direction of the diode in the circuit of Figure 5. Then, observe and record the output voltage V_o . How does the output voltage compare to that of the last step? **(2 Points)**



The clipping on the bottom half is much sharper than our previous plot.

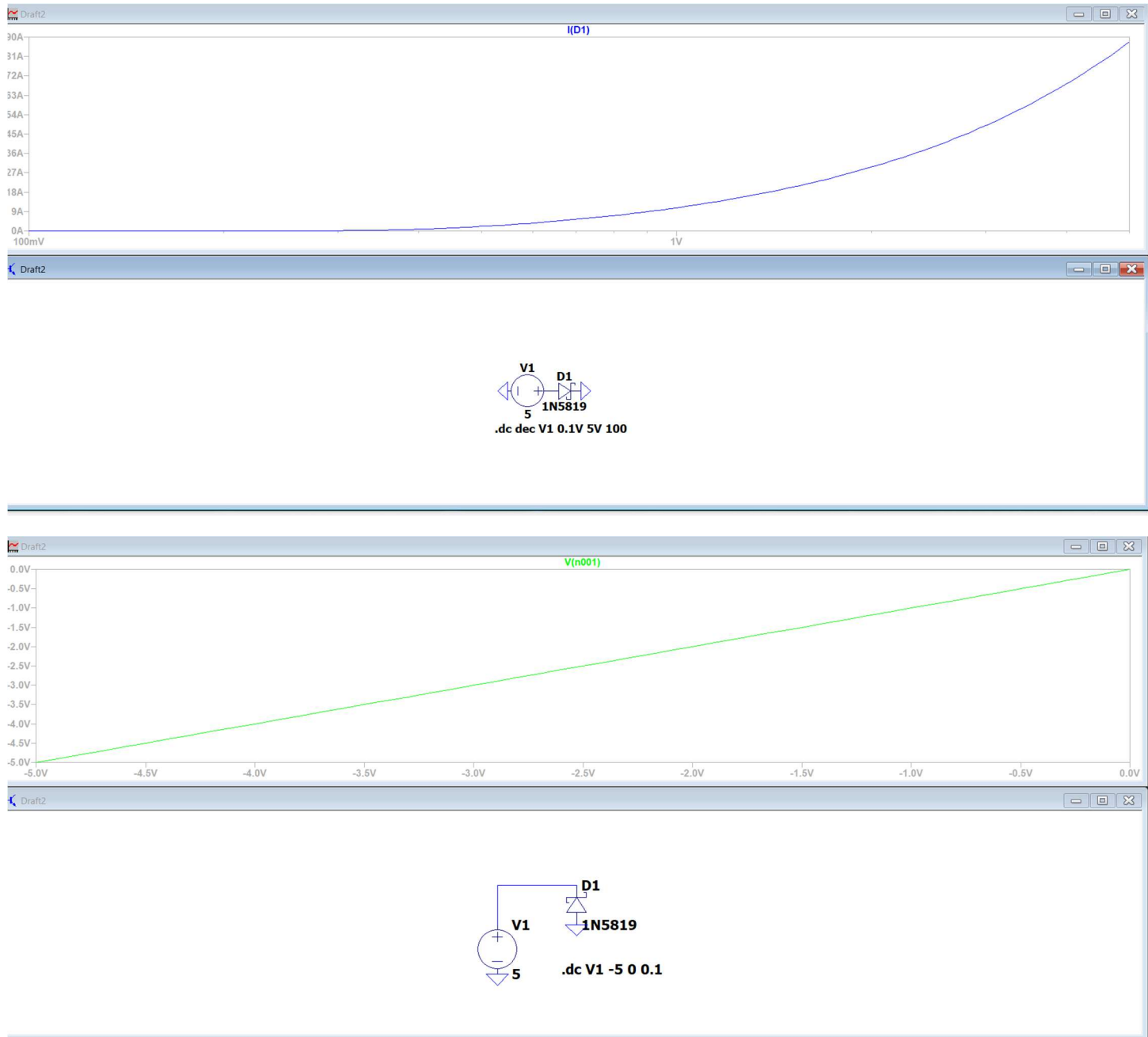
- VII. Construct the circuit shown in Figure 6. Let $V_i(t) = 5.0\sin(377t)$ (frequency = 60 Hz), $V_{r1} = 3.0$ V, and $V_{r2} = 2.0$ V. Provide the output waveform $V_o(t)$ and show the maximum and minimum values of the output voltage. **(4 Points)**



Part 5: Schottky Diode Characteristics (10 Points)

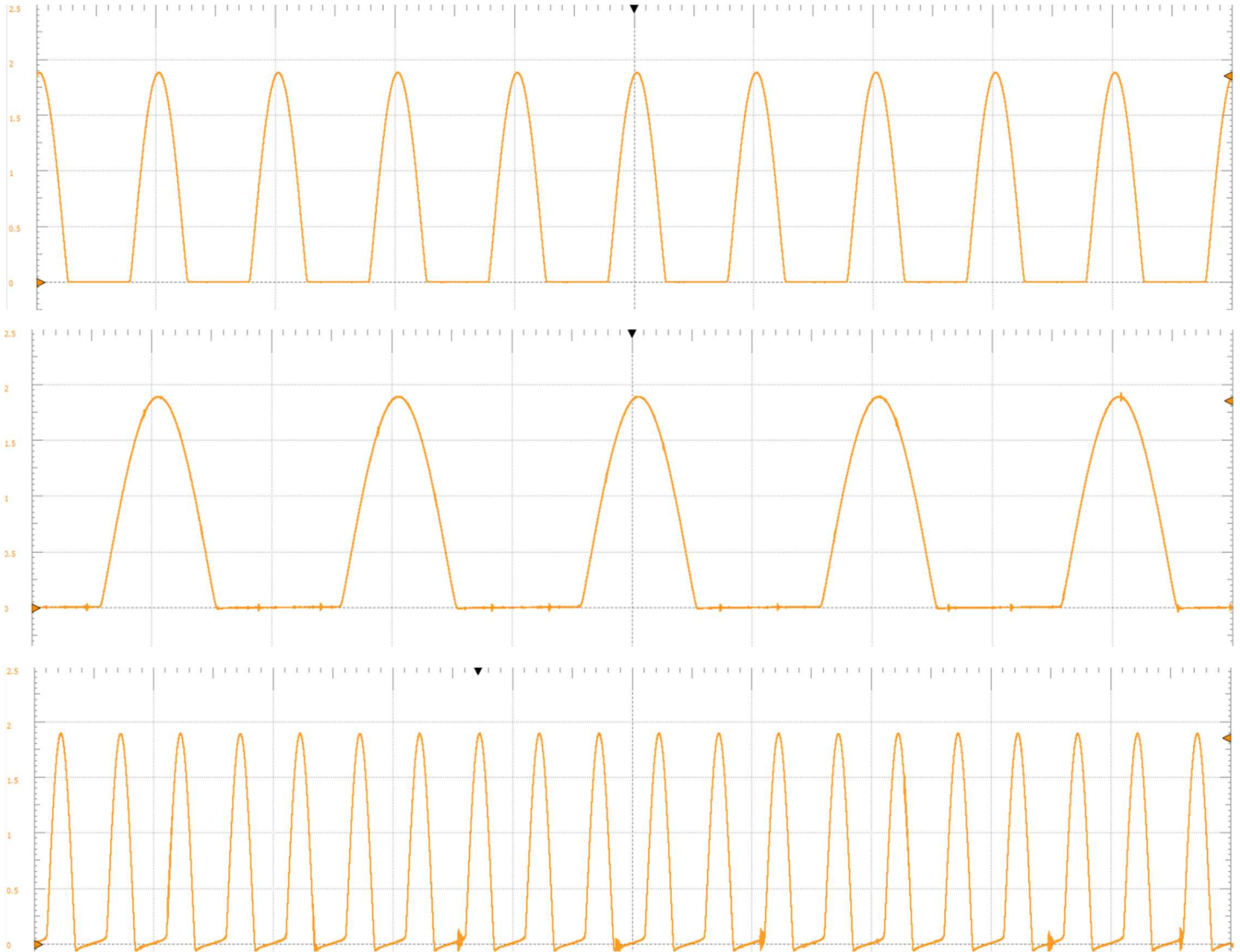
Schottky diodes are also an important diode that you should be familiar with in analog circuitry. Be sure to understand the differences between these various diodes when building circuits as each of them have their own unique properties that make them advantageous for certain projects.

- I. Create a circuit in LTspice with a 1N5822 Schottky diode in forward bias with a voltage source. Run a simulation to obtain the IV curve of the diode with a voltage range of 0 V to 5 V. Run another simulation obtaining the IV curve from -5 V to 0 V, except this time, place the diode in reverse bias (flip the polarity). Determine the cut-in voltage of the diode. **(5 Points)**



- II. Now connect a 5 V_{pp}, 1 kHz sine wave input to the half wave rectifier circuit including Schottky diode and a series resistor of 75 Ω . Take a screenshot of this waveform. Increase the frequency to

higher values and note exactly till what frequency the Schottky diode rectifier is able to maintain its rectifying property. **(5 Points)**

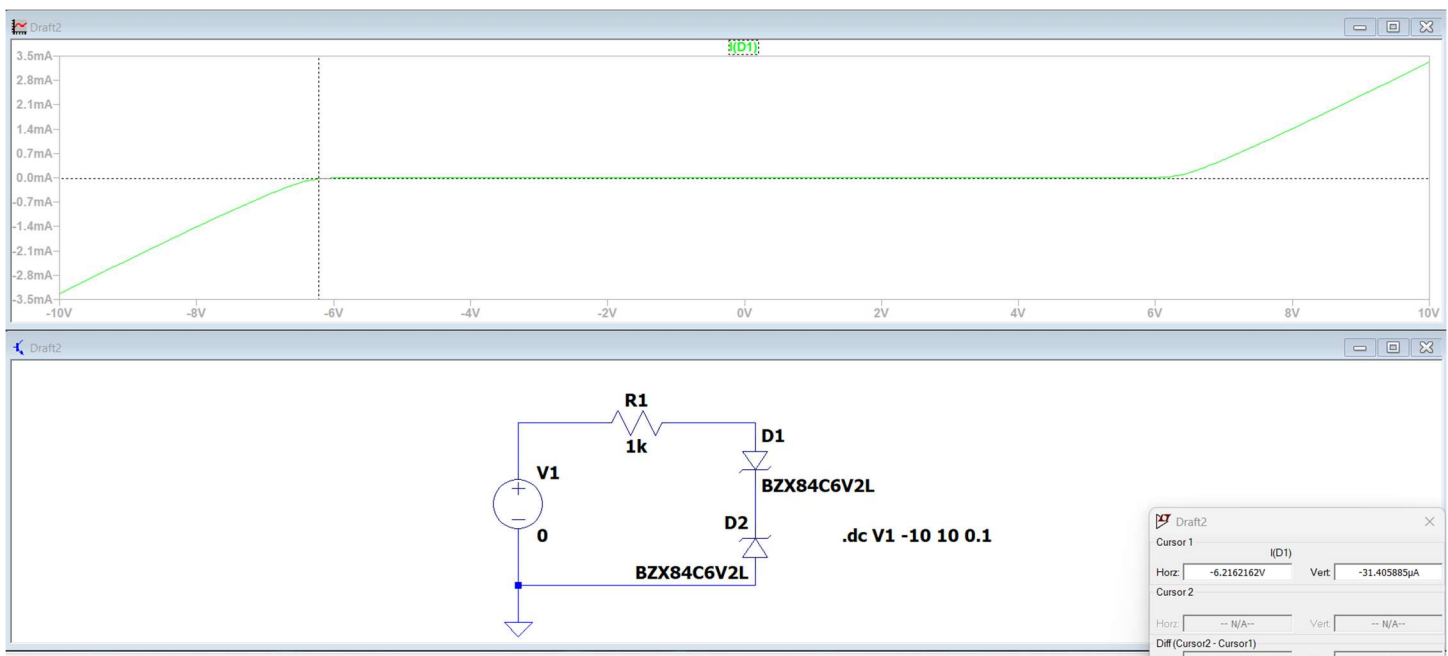
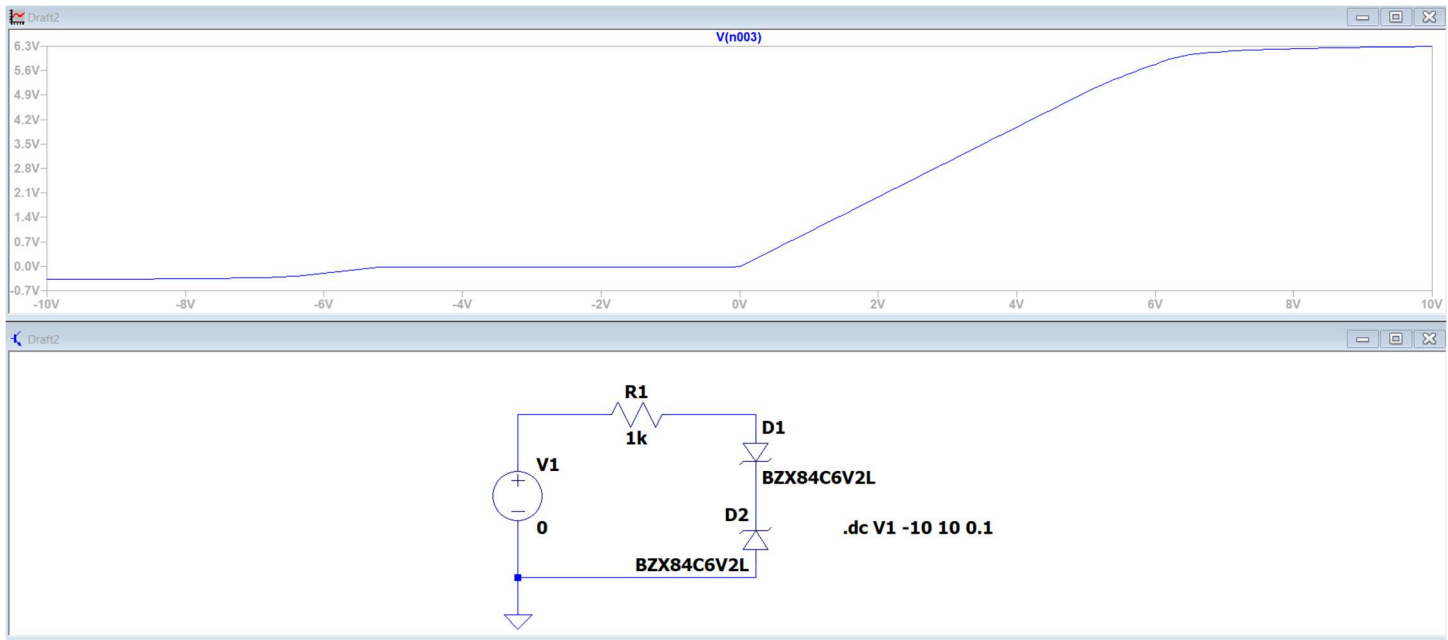


We see it rectifies up until we push it to around ~100kHz

Part 6: Zener Diode Characteristics / Zener Clamp (10 Points)

The following circuit demonstrates similar outputs of previous circuits except this time we can use zener diodes instead to achieve this V_{out} in between the two Zener diodes.

- I. Create the circuit in LTspice with a BZX84C6V2L in forward bias with a voltage source. Run a simulation to obtain the IV curve of the diode with a voltage range of -10 V to 10 V. Determine the reverse breakdown voltage of the diode. (5 Points)



-6.2V gives our breakdown

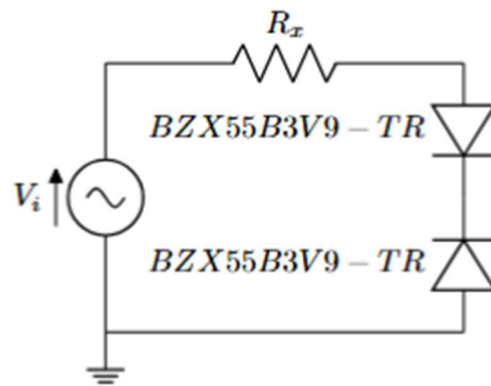


Figure 7 – Zener Diode Clamp

- II. Now we are going to construct a Zener-Based Clamp Circuit. Construct the following circuit with your Zener diodes and capture the effect of the circuit at the output between the two diodes on your DAD scope. Use $R_x = 1 \text{ k}\Omega$ and $V_i = 2 \text{ V}$, 1 kHz sine wave. **(5 Points)**

